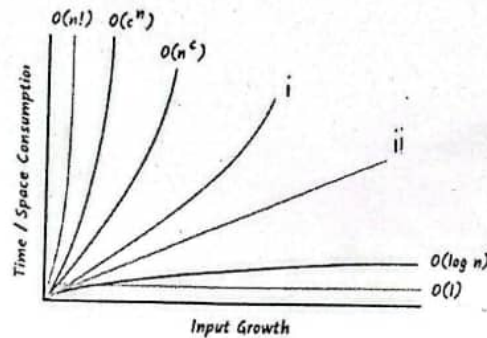


Bismillahir Rahmanir Rahim
International Islamic University Chittagong
 Department of Computer Science & Engineering
Mid Term Examination, Autumn 2025
CSE 2321 Data Structures
 Total marks: 30 Time: 90 minutes

[Answer *all* of the following questions. Figures in the right-hand margin indicate full marks.]

1. a) Explain why Big O notation is preferred over measuring algorithm time in seconds.
 Find the Time complexity of (i) and (ii) on the following graph.

1
+
2



- b) Draw a flowchart or write a code for linear search algorithm.

2

[OR]

Draw a flowchart or write a code for finding the second largest element of an array.

- c) (i) Given the pattern $P = \text{'aac'}$ and the text $T = \text{'abaaaccacaaaexx'}$, determine the number N of character comparisons required to find the first occurrence (index) of P in T using the naïve pattern matching algorithm.

3
+

Show the matching process step by step until the pattern is found.

1
+

- ii) State the time complexity of the pattern-matching algorithm you used.

1

- iii) Give two pairs of examples of pattern P and text T , where, pattern length $|P| = 4$ and text length $|T| = 10$:

- One pair gives the minimum number of character comparisons.
- One pair gives the maximum number of character comparisons.

2. a) Explain how to represent a two-dimensional array in computer memory using row-major order and column-major order. Illustrate your answer with an appropriate example or diagram.

3

- b) Consider a 2D array $A(4:12, 7:14)$.

3

- i. Find the length of each dimension and the total number of elements in A .
- ii. If $\text{Base}(A) = 300$ and $w = 8$ bytes per memory cell, calculate the address of the element $A[6, 10]$ assuming row-major storage order.

- c) Given a string **CANDLE**, find the number of comparisons (C) and number of interchanges (D) needed to sort the string alphabetically by using bubble sort algorithm. Show each steps.

4

[OR]

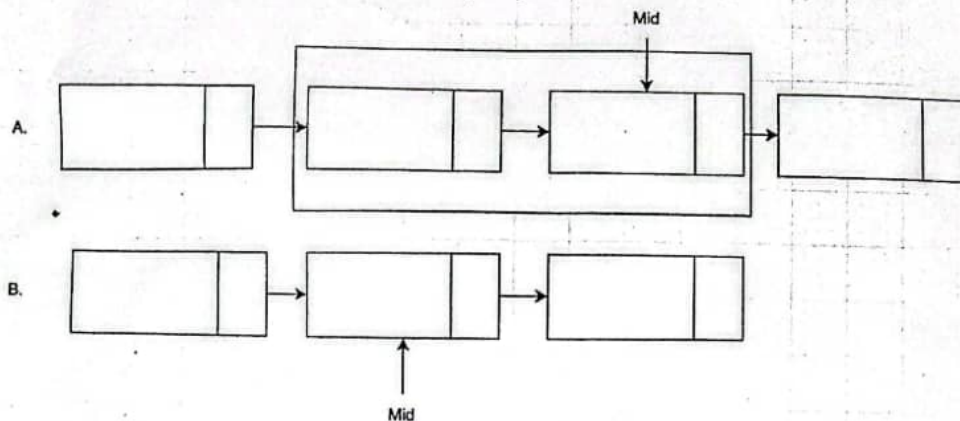
What is the time complexity of the binary search algorithm?

Show the intermediate steps to search the $\text{ITEM} = 41$ and $\text{ITEM} = 97$ from the following list of integers using binary search algorithm.

21, 25, 30, 35, 41, 45, 53, 66, 68, 71, 82, 89, 101

3. a) What is the difference between Array and Linked List? Which one is better for searching an element and why? 2

b) (i) Write an algorithm or a program to insert a node in the middle position of a Linked List. Your algorithm/program should handle the following cases: 3
 1. When node number is even, return as stated in A. +
 2. When node number is odd, return as stated in B. 1



(ii) Explain how the same insertion can be achieved by traversing the linked list only once.

[OR]

You are managing a reading list using a doubly linked list. Each node stores a book title. The list has 5 books with the following addresses: 2
 +
 1
 +
 1

Book Title	Node Address	Previous Address	Next Address
Book C	1502	3071	4096
Book A	2005	NULL	3071
Book E	2810	4096	NULL
Book B	3071	2005	1502
Book D	4096	1502	2810

Tasks:

1. Draw the doubly linked list showing nodes, data, previous and next pointers.
2. Insert a new book "Book F" after "Book C" and show the updated links.
3. Remove "Book B" from the reading list and update the links accordingly.

c) Write an algorithm or program to insert a node into a singly linked list at a given position (pos, starting from 1). Your solution should: 1
 +

- Insert in $O(1)$ time if inserting at the beginning or end 2
- Insert in $O(n)$ time when inserting at any intermediate position. +
- Handle invalid positions appropriately. 1

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Mid Term Examination, Spring 2025
CSE 2321 Data Structures
 Total marks: 30 Time: 90 minutes

[Answer *all* of the following questions. Figures in the right-hand margin indicate full marks.]

1. a) What do you mean by a *data structure*? Briefly describe the different operations performed on data structures. 2 CO1
C2

- b) Draw a *flowchart* for *linear search* algorithm. 3 CO1
C2
 [Or]
 Draw a *flowchart* to find the *smallest element* of an array.

- c) What is the *time complexity* of the following functions in Big O notation? Explain. 2 CO3
C4

i) <pre>int solve1(int n){ int i = n; int cnt = 0; while(i>0){ i = i / 2; cnt++; } return cnt; }</pre>	ii) <pre>int solve2(int n){ int cnt = 0; for(int i=0; i<n; i++){ for(int j=1; j<n; j*=2){ cnt = cnt + 1; } } return cnt; }</pre>
--	---

- d) Given the pattern **P = 'ccd'** and text **T = 'cbcdcdcdcdcccd'**, determine the number **N** of character comparisons required to find the first occurrence (index) of **P** in **T** using the naive *pattern matching algorithm*. Show the matching process step by step until the pattern is found. 3 CO3
C3

2. a) Write an algorithm to *insert* an element **ITEM** to the **Kth** position of a linear array **LA**. What is the time complexity of this algorithm? 2 CO1
C2

- b) Consider a 2D array **A(4:12, 7:14)**. 2.5 CO3
C3
 - i. Find the length of each dimension and the total number of elements in **A**.
 - ii. If **Base(A) = 500** and **w = 4** bytes per memory cell, calculate the address of the element **A[6, 10]** assuming row-major storage order.

- c) Given a string **BRIDGE**, find the number of *comparisons* (**C**) and number of *interchanges* (**D**) needed to *sort* the string *alphabetically* by using *bubble sort* algorithm. Show each steps. 4 CO3
C3
 [OR]
 You are developing a search feature for a library management system that stores book IDs in a sorted list. To enable quick look-ups, which search algorithm would you choose to efficiently check if a given book ID exists? Write the algorithm for this search.

- d) Write down the *worst case* & *average case* time complexity for the following algorithm: 1.5 CO1
C1
 - i) Binary search ii) Linear search iii) Bubble Sort

3. a) What is *linked list*? What is the difference between an array and a linked list in terms of their memory structure and operations? Which one is more efficient for searching an element, and why? 2 CO1 C1
- b) Describe how you can *insert* a new node at the middle position of a singly linked list. Write an **algorithm or C/C++ function** for the insertion, and state the time complexity of this insertion operation. 4 CO3 C3

[OR]

You are developing a **music playlist manager** using a **singly linked list**. Each node in the list stores the name of a song. The playlist contains an **odd number of songs**, but the number of songs is **unknown**. Write an **efficient algorithm or C/C++ function** to find the **middle song** of the playlist using **only one traversal**.

- c) Given is the following sorted linked list where the array **INFO** contains a list of integer numbers and **LINK**, **START** and **AVAIL** are the pointer fields- 2 CO1 C2

	1	2	3	4	5	6	7	8	9	10	11	12
INFO:			41	129		9	123	78	194	231	62	145
LINK:	2	5	11	12	0	3	4	7	10	0	8	9

START : 6 AVAIL : 1

Determine the changes in the list if number **XX** is *added* to the list and then **145** is *deleted* from the list.

[Here **XX** is the last two digits of your ID]

- d) What is *two way linked list*? Form a *two way linked list* from the one way list in Q 3(c). 2 CO1 C3

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 Department of Computer Science & Engineering
Mid Term Examination, Spring 2024
CSE 2321 Data Structures
 Total marks: 30 Time: 90 minutes

[Answer all of the following questions. Figures in the right-hand margin indicate full marks.]

1. a) A professor keeps a class list containing the following data for each student: 2. CO1 C2
 Name, ID Number, Section, Total Marks, Final Grade
 i) State the *entities, attributes* and *entity set* of the list.
 ii) Which attribute can serve as the *primary key* for the list?
- b) Draw a flowchart for binary search algorithm 3 CO1 C2
 Or
 Draw a flowchart to determine whether a given number *N* is *prime* or not.
- c) What do you mean by *space complexity*? What is the *space complexity* of the following 2 CO2 C2
 functions? Explain.
 i)

```
void swap()
{
    int a, b, temp;
    temp = a;
    a = b;
    b = temp;
}
```


 ii)

```
int sum(int a[], int n)
{
    int i, sum = 0;
    for(i = 0; i < n; i += 2)
        sum += a[i];
    return sum;
}
```
- d) For the following pattern *P* and text *T*, find the number *C* of comparisons to find the **INDEX** 3. CO3 C3
 of *P* in *T* using the *pattern matching algorithm* you studied. You have to show each step.
P = ccd, *T* = cbcddcdddcddcccccdd
2. a) Write an algorithm to *insert* an element *ITEM* to the *Kth* position of a linear array *LA*. What 2.5 CO1 C2
 is the time complexity of your algorithm?
- b) Consider an array *B*(1:8, -5:5, -10:5). 2.5 CO3 C3
 i) Find the *length* of each dimension and the *number of elements* in *B*.
 ii) Suppose *Base (B)* = 150 and *w* = 4 words per memory cell for *B*.
 Find the effective indices *E₁*, *E₂*, *E₃* and the address of the element *B* [2, -3, 2] assuming *B*
 is stored in *row-major order*.
- c) Given a string **STRUCTURES**, find the number of *comparisons* (*C*) and number of 3.5 CO3 C3
interchanges (*D*) needed to *sort* the string *alphabetically* by using *bubble sort* algorithm.
 Show each steps.
 Or
 Modify the *bubble sort* algorithm so that it will stop early if it detects that the list is already
 sorted. Write the modified *bubble sort* algorithm.
- d) Write down the *worst case* & *average case* complexity for the following algorithm: 1.5 CO1 C1
 i) Binary search ii) Linear search iii) Bubble Sort

3. a) What is *stack*? Write a procedure to **PUSH** an item onto a stack and **POP** an item from the stack. 2.5 CO1 C2

b) Suppose the following stack of integers is in memory where **STACK** is allocated $N = 8$ memory cells. 1 CO1 C2

TOP = 4 **STACK:** 11, 22 33, 44, __, __, __, __

Describe the **STACK** and **TOP** as the following operations take place –

- i) POP (STACK, ITEM) iii) PUSH (STACK, 99)
- ii) PUSH (STACK, 77) iv) POP (STACK, ITEM)

c) Consider the following postfix expression, 2.5 CO3 C2

P: 3, 2, +, 2, ^, 7, 4, -, 2, *, +, 5, -

Evaluate P, using the algorithm you have studied.

d) Consider the following *infix* expression Q: 4 CO3 C2

Q: $((A + B) / D) ^ ((E - F) * G)$

Translate Q into its equivalent *postfix* expression P using the algorithm you studied.

Or

Write an algorithm that receives a bracket sequence and tells whether the sequence is correct or incorrect.

For example, "()", "()[]{ }", "({ })[]" are correct sequences while, "({ })[])", "{]" are not.

[You can safely assume that the required data structure and its operational algorithms are available to you to solve this problem. You don't need to re-implement them.]

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Mid Term Examination, Autumn 2023
CSE 2321 Data Structures
Total marks: 30 Time: 90 minutes

[Answer *all* of the following questions. Figures in the right-hand margin indicate full marks.]

- 1.a) A company maintains an employee database with the following data for each employee: 2 CO1
Employee ID, Name, Department, Salary, Joining Date
i) State the *entities*, *attributes* and *entity set* of the list.
ii) Which attribute can serve as the *primary key* for the list?
- b) Draw a *flowchart* for the *binary search* algorithm. 3 CO1
OR
Draw a *flowchart* for the *bubble sort* algorithm.
- c) What do you mean by *complexity of algorithms*? What is the *time complexity* of the following functions? Explain. 2.5 CO2
i)

```
void fun1 (int N)
{
    int a = 0, i = N;
    do {
        a += i;
        i /= 2;
    } while (i > 0);
}
```

ii)

```
int fun2 (int n)
{
    int i, j, sum = 0;
    for(i = 0; i * i < n; i++) {
        for (j = 1; j * j <= n; j++) {
            sum += i * j;
        }
    }
    return sum;
}
```
- d) Suppose, S1 = "GLADIATOR", S2 = "DESH", S3 = "BAN". Evaluate and write the output of the following operations in order [Assume first Index of the string is 1]: 2.5 CO2
i) L = LENGTH (S3)
ii) S4 = DELETE (S1, 4, L + 3)
iii) S5 = S3 || S2
iv) S5 = INSERT (S5, 4, S4)
v) INDEX (S5, "NGL")
- 2.a) What is *linear array*? How can we represent a linear array in memory? 1 CO1
OR
What is *sparse matrix*? Describe how memory can be efficiently utilized by using a sparse matrix representation.
- b) Consider a 2D array A (5 : 10, 8 : 15). 2 CO3
i) Find the length of each dimension and the number of elements in A.
ii) Suppose **Base (A) = 200** and **w = 4** words per memory cell for A.
Find the address of the element A [7, 10] in *row-major order*.
- c) Let A be $n \times n$ square matrix. Write a module which 3 CO4
i) Find the number NUM of *zero* elements in A.
ii) Find the SUM of the elements above the diagonal i.e. elements A[i, J] where $i < J$.
- OR
Modify the *bubble sort* algorithm so that it will stop early if it detects that the list is already sorted.

- d) Show the steps to search the ITEM = X and ITEM = Y from the following list of integers using *binary search* algorithm. 4
- 11, 22, 30, 35, 41, 45, 53, 66, 68, 71, 82, 89, 99**
- [Here, X is the first two digits and Y is the last two digits of your ID. For example, if ID = C161026, then X = 16 and Y = 26]

- 3.a) Suppose STACK is allocated N = 8 memory cells and initially STACK is empty (i.e. TOP = 0). 2 CO1
Find the output of the following module [show the elements of STACK and value of TOP in each step]-

1. Set A := 15 and B := 17
2. Call PUSH (STACK, A)
Call PUSH (STACK, 5)
Call PUSH (STACK, B - 3)
Call PUSH (STACK, 10)
Call PUSH (STACK, A * B)
Call PUSH (STACK, 9)
3. Repeat while TOP $\neq$ 0:
Call POP (STACK, ITEM)
Write: ITEM
[End of loop]
4. Return

- b) Write an algorithm/code to *reverse* the elements of a stack. 2.5 CO3

- c) Given a postfix expression: 2.5 CO3

Q: 9, -4, +, 3, $\uparrow$, N, 4, -, 2, *, +, 5, -

For what value of N, the result of the expression Q would be XY.

[Here, XY is the last two digits of your ID. For example, if ID = C161026, then XY = 26. The result of any division operator must be taken 2 places after the decimal point. For example, $26 / 5 = 5.20$]

- d) Consider the following *infix* expression Q: 3 CO3

Q: $A * (B - D) \uparrow E + F * (G - H / K)$

Translate Q into its equivalent *postfix* expression P using the algorithm you studied.

OR

Suppose a stack supports the following operations in constant time complexity ($O(1)$):

push(x): Pushes an element x onto the stack.
pop(): Removes the element on top of the stack.
top(): Retrieves the element on top of the stack.
getMin(): Retrieves the minimum element in the stack.

Now write algorithms/code of the push(x) and getMin() functions for this special stack

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 Department of Computer Science & Engineering
Mid Term Examination, Spring 2023
CSE 2321 Data Structures
 Total marks: 30 Time: 90 minutes

[Answer all of the following questions. Figures in the right-hand margin indicate full marks.]

1. a) A library maintains a book list containing the following data for each book:
 Name of Authors, Title of Book, Edition, Publisher's Name, Year, ISBN
 i) State the *entities, attributes* and *entity set* of the list.
 ii) Which attribute can serve as the *primary key* for the list?
 2 CO1
- b) Draw a flowchart for linear search algorithm
 Or
 Draw a flowchart to find the largest element of an array.
 3 CO1
- c) What is the time complexity of the following functions? Explain.
 2 CO2
- i)
 void swap()
 {
 int a, b, temp;
 temp = a;
 a = b;
 b = temp;
 }

iii)
 int add3(int n)
 {
 int i, sum = 0;
 for(i = 1; i < n; i *= 2)
 sum += i;
 return sum;
 }

ii)
 int add2(int n)
 {
 int i, sum = 0;
 for(i = 0; i < n; i += 2)
 sum += i;
 return sum;
 }

iv)
 int fun(int n)
 {
 int count = 0;
 for (int i = n; i > 0; i--)
 for (int j = 0; j < i; j++)
 count += 1;
 return count;
 }
- d) For the following pattern P and text T, find the number C of comparisons to find the INDEX of P in T using the *pattern matching algorithm* you studied. You have to show each step.
 P = aab, T = ababbabbbabaaaabbb
 3 CO3
2. a) Suppose ACAD maintains a linear array STUCGPA(23001:23099) such that STUCGPA[K] contains the CGPA of the student who bears the ID number K. Write a module for each of the following tasks-
 2 CO3
- i) Print the ID Number of the students who have CGPA above 3.90.
 ii) Count the number of students who have CGPA above 3.30.

3 CO3
- b) Consider an array A (3:10, 1:10, -3:10).
 i) Find the *length* of each dimension and the *number of elements* in A.
 ii) Suppose Base (A) = 200 and w = 4 words per memory cell for A.
 Find the effective indices E₁, E₂, E₃ and the address of the element A [8, 6, 4] assuming A is stored in *row-major order*.
 4 CO3
- c) Given a string CHITTAGONG, find the number of comparisons (C) and number of interchanges (D) needed to sort the string *alphabetically* by using *bubble sort* algorithm. Show each steps.
 Or
 Modify the *Binary Search algorithm* so that it becomes a search and insertion algorithm.
 1 CO1
- d) What is the main i) advantage and ii) disadvantage of the Binary Search algorithm over the Linear Search algorithm?

3 a) Write a procedure to PUSH an item onto a stack and POP an item from the stack.

b) Suppose the following stack of integers is in memory where STACK is allocated $N = 8$ memory cells:

TOP = 4 STACK: 11, 22 33, 44, __, __, __, __

Describe the stack and TOP as the following operations take place –

- i) POP (STACK, ITEM) iii) PUSH (STACK, 66)
- ii) PUSH (STACK, 55) iv) POP (STACK, ITEM)

c) Consider the following postfix expression,

P: 3, 1, -, 2, ↑, N, 4, /, 2, *, +, 5, -

For which value of **N**, the result of this expression would be **XY**, where **XY** is the last 2 digits of your ID? [If your ID is C161026, then **XY** = 26] Use the algorithm you studied to solve this problem.

d) Consider the following *infix* expression Q:

Q: $(A * B \uparrow D) / (E - F) + G$

Translate Q into its equivalent *postfix* expression P using the algorithm you studied.

Or

Write an algorithm that receives a bracket sequence and tells whether the sequence is correct or incorrect.

For example, "()", "()[]{ }", "{ }[]" are correct sequences while, "{ }[])", "{ }" are not. [You can safely assume that the required data structure and its operational algorithms are available to you to solve this problem. You don't need to re-implement them.]

[Answer **all** of the following questions. Figures in the right-hand margin indicate full marks.]

- | Sl. No. | Question | Mark | CO | PO |
|---------|--|------|-----|----|
| 1. a) | A professor keeps a class list containing the following data for each student:
Name, ID Number, Section, Total Marks, Final Grade
i) State the <i>entities, attributes</i> and <i>entity set</i> of the list.
ii) Which attribute can serve as the <i>primary key</i> for the list? | 2 | CO1 | C2 |
| b) | Draw a <i>flowchart</i> for the <i>binary search</i> algorithm.
OR
Draw a <i>flowchart</i> for the <i>bubble sort</i> algorithm. | 3 | CO1 | C2 |
| c) | What do you mean by <i>complexity of algorithms</i> ? Analyze the complexity of the <i>linear search algorithm</i> . | 2.5 | CO2 | C2 |
| d) | For the following pattern P and text T, find the number C of comparisons to find the INDEX of P in T using the <i>pattern matching algorithm</i> you studied. You have to show each step.
P = abba, T = <u>abaabaaabbbbaabbab</u> | 2.5 | CO3 | C3 |
| 2. a) | What is <i>linear array</i> ? How can we represent a linear array in memory?
OR
How a 2D <i>array</i> is represented in computer memory? Explain with example. | 1 | CO1 | C2 |
| b) | Consider a 2D array A (5 : 10, 8 : 15).
i) Find the length of each dimension and the number of elements in A.
ii) Suppose Base (A) = 200 and w = 4 words per memory cell for A.
Find the address of the element A [8, 12] in <i>row-major order</i> . | 2 | CO3 | C3 |
| c) | Let A be n x n square matrix. Write a module which
i) Find the number NUM of zero elements in A.
ii) Find the SUM of the elements above the diagonal i.e. elements A[i, j] where $i < j$. | 3 | CO4 | C5 |

- d) Given a string **CONVOCATION**, find the number of comparisons (C) and number of interchanges (D) needed to sort the string *alphabetically* by using *bubble sort* algorithm. Show each step. -4 CO3 C1
- OR
- Show the intermediate steps to search the **ITEM = 41** and **ITEM = 97** from the following list of integers using *binary search* algorithm.
- 21, 25, 30, 35, 41, 45, 53, 66, 68, 71, 82, 89, 101
3. a) What is *stack*? Write a procedure to **PUSH** an item onto a stack and **POP** an item from the stack. 2.5 CO1 C2
- OR
- If **P** be an arithmetic expression written in *postfix expression*, write an algorithm to evaluate **P**, using a stack to hold operands
- b) Suppose **STACK** is allocated **N= 6** memory cells and initially **STACK** is empty (i.e. **TOP=0**). Find the output of the following module [show the elements of **STACK** and value of **TOP** in each step]- 2 CO1 C2
1. Set **AAA := 10** and **BBB := 20**
 2. CALL **PUSH(STACK, AAA)**
CALL **PUSH(STACK, 5)**
CALL **PUSH(STACK, BBB+3)**
CALL **PUSH(STACK, 15)**
CALL **PUSH(STACK, AAA+BBB)**
 3. Repeat while **TOP ≠ 0**:
CALL **POP(STACK, ITEM)**
Write: **ITEM**
[End of loop.]
 4. Return.
- c) Consider the following arithmetic expression **P**, written in *postfix* notation: 2.5 CO3 C3
- P: 20, 7, 3, -, /, 5, 2, 8, -, *, +**
- Evaluate **P**, using the algorithm you have studied.
- d) Consider the following *infix* expression **Q**: 3 CO3 C3
- Q: A * (B - D) ↑ E + F * (G - H / K)**
- Translate **Q** into its equivalent *postfix* expression **P** using the algorithm you have studied.

Set: Alpha
International Islamic University Chittagong
Department of Computer Science & Engineering
Mid Term Examination, Spring 2022
CSE 2321 Data Structures

Total marks: **21**

Time: 2 hours 30 minutes for exam + 30 minutes for submission

[Answer **all** of the following questions. Figures in the right-hand margin indicate full marks.]

- | | | | |
|---|---|-----|----|
| 1. a) A library maintains a book list containing the following data for each book:
Name of Authors, Title of Book, Edition, Publisher's Name, Year, ISBN
i) State the <i>entities</i> , <i>attributes</i> and <i>entity set</i> of the list.
ii) Which attribute can serve as the <i>primary key</i> for the list? | 1 | CO1 | C2 |
| b) Draw a <i>flowchart</i> for <i>linear search</i> algorithm [if last digit of your ID is odd] / <i>binary search</i> algorithm [if last digit of your ID is even]. | 2 | CO1 | C2 |
| c) What is the <i>time complexity</i> of the following functions? Explain. | 2 | CO2 | C4 |
- i)

```
void swap()
{
    int a, b, temp;
    temp = a;
    a = b;
    b = temp;
}
```

iii)

```
void fun2(int N)
{
    int a = 0, i = N;
    while (i > 0) {
        a += i;
        i /= 2;
    }
}
```

ii)

```
int fun1(int N)
{
    int i, j, a = 0;
    for (i = 0; i < N; i++) {
        for (j = N; j > i; j--) {
            a = a + i + j;
        }
    }
    return a;
}
```

iv)

```
int fun(int n)
{
    int count = 0;
    for (int i = n; i > 0; i--)
        count += 1;
    return count;
}
```

- d) For each of the following patterns P and texts T, find the number C of comparisons to find the INDEX of P in T using the *pattern matching algorithm* you studied. 2 C03 C3
- i) P = aab, T = ababbabbbbbaaaabbbb
 - ii) T = 1234567890, P = X (X is the last two digits of our ID, i.e. if your ID is C193231, then P=X=31)
2. a) Suppose ACAD maintains a linear array STUCGPA(22001:22099) such that STUCGPA[K] contains the CGPA of the student who bears the ID number K. Write a module for each of the following tasks- 1.5 CO4 C5
- i) Print the ID Number of the students who have CGPA above 3.75.
 - ii) Count the number of students who have CGPA above 3.50.
- b) Consider an array A (5:10, 1:10, -2:10). 2 C03 C3
- i) Find the *length* of each dimension and the *number of elements* in A.
 - ii) Suppose **Base (A) = 20Y** (Y is the last digit of our ID, i.e. if your ID is C193207, then Y=7) and w = 4 words per memory cell for A.
Find the effective indices E₁, E₂, E₃ and the address of the element A [7, 5, -1] assuming A is stored in *row-major order* [if last digit of your ID is even] / *column-major order* [if last digit of your ID is odd]
- c) Given a string **MORALITY**, find the number of *comparisons* (C) and number of *interchanges* (D) needed to *sort* the string *alphabetically* by using *bubble sort* algorithm. 1.5 CO3 C3
- d) Each student in a class of 40 students takes 5 tests in which scores range between 0 and 100. Suppose the test scores are stored in a 40x5 array TEST. Write a module which 2 CO4 C5
- i) Find the average mark for each test.
 - ii) Finds the final grade for each student where the final grade is the average of the student's three highest test scores.

3. a) Suppose the following stack of integers is in memory where STACK is allocated N = 8 memory cells:

TOP = 4 STACK: 11, 22, N, 33, __, __, __, __

Describe the stack as the following operations take place –

- i) POP (STACK, ITEM) iv) PUSH (STACK, 66)
- ii) PUSH (STACK, 44) v) POP (STACK, ITEM)
- iii) PUSH (STACK, 55) vi) POP (STACK, ITEM)

[Here N is the last two digits of our ID, i.e. if your ID is C193216, then N=16]

- b) Suppose stacks A[1] and A[2] are stored in a linear array STACK with N elements, as pictured in Fig. 6.37. Assume TOP[K] denotes the top of stack A[K].

- (a) Write a procedure PUSH(STACK, N, TOP, ITEM, K) which pushes ITEM onto stack A[K].
- (b) Write a procedure POP(STACK, TOP, ITEM, K) which deletes the top element from stack A[K] and assigns the element to the variable ITEM.

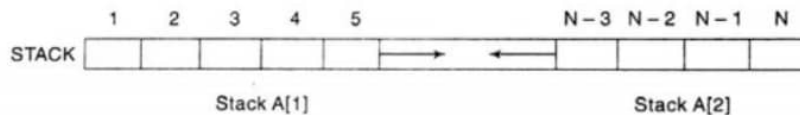


Fig. 6.37

- c) Evaluate each of the following postfix expression using the algorithm that you have studied.

P₁: N, 7, 3, +, *, 5, 2, 8, -, /, +
P₂: 3, 1, +, 2, ↑, N, 4, -, 2, /, +, 5, *

[Here, N is the last digit of your ID. For example, if ID = C193009 then N = 9]

- d) Consider the following *infix* expression Q:

Q: (A * (B - D) ↑ E + F) * (G - H / K)

Translate Q into its equivalent *postfix* expression P using the algorithm you studied.